EMILLY RAIANE RODRIGUES

emillyraianerodrigues@gmail.com +55 55 9 9650 1373
linkedin.com/in/emilly-raiane- github.com/Raiagues Santa Maria, Brazil

EDUCATION

Bachelor of Engineering in Aerospace Engineering **Expected Dec. 2026**
Federal University of Santa Maria (UFSM) *Santa Maria, Brazil*

- Selected as one of 43 from 11,000+ applicants for Líderes Estudar based on academic excellence and leadership potential.
- Won the South Region Prêmio Na Prática for leadership and impact, selected from 8,000+ applicants.
- Selected as 1 of 10 Brazilian students for a fully funded innovation conference at Google in Sunnyvale.
- Spoke on two New Space panels (200+ attendees each) at South Summit Brazil (23,000+ participants; 900+ investors).

Master of Engineering in Embedded Systems and Connected Objects – Dual Degree **Sep. 2025**
Grenoble INP - Phelma *Grenoble, France*

- Awarded a two-year CAPES-BRAFITEC merit scholarship for academic performance to complete the dual degree in France.

EXPERIENCE

Machine Learning Research Intern **Jul. 2026–Sep. 2026**
European Organization for Nuclear Research (CERN) *Geneva, Switzerland*

- Selected as one of 47 students from nearly 6,000 applicants worldwide for CERN's openlab program.
- Developed an AI model for ATLAS to reconstruct particle events from detector data in real time.
- Designed a custom model with fewer than 800 parameters and began deploying it under strict hardware limits.
- Reached 81% detection efficiency and 80% purity with noisy inputs, and 93% for both without noise.

Satellite Embedded Flight Software Intern **Oct. 2025–Jun. 2026**
Visiona Space Technology *São José dos Campos, Brazil*

- Developed C flight software for a nanosatellite across commands, telemetry, imaging, and spacecraft modes.
- Implemented location logic that started imaging sequences about 30 seconds before target regions.
- Coordinated camera, antenna, communications, equipment, and attitude-control commands into mission sequences.
- Tested ground commands and telemetry end to end and reviewed requirements and interfaces across subsystem teams.
- Recreated build and test environments with Docker and automated software integration through GitLab.

Satellite Embedded Flight Software Intern **Feb. 2025–Aug. 2025**
French Space Agency (CNES) *Toulouse, France*

- Developed real-time C flight software for recovery, task scheduling, spacecraft modes, and power states.
- Designed a reset-recovery mechanism that cut satellite downtime from seven hours to minutes.
- Moved non-critical memory reads to background tasks, reducing startup blocking from 40+ seconds to under 5.
- Built and debugged a low-level driver for persistent memory connected through programmable hardware.
- Ran hardware-in-the-loop tests and automated configuration, builds, and system tests with Python and Bash.

Fault-Tolerant Embedded Software Intern **May 2024–Aug. 2024**
TIMA Laboratory *Grenoble, France*

- Developed low-level C communication between an STM32 and Raspberry Pi for radiation experiments.
- Configured hardware registers, clocks, interrupts, and communication drivers directly on the STM32.
- Reduced a verification step from 8.6 seconds to 0.014 seconds by tracking communication state.
- Made interrupted experiments restartable and automated test control, error handling, and data collection.

PROJECTS

Interactive NASA Flight Software Platform [GitHub] **Sep. 2025–Present**

- Building a browser platform that helps students learn real satellite flight software without difficult setup.
- Designed for universities and student satellite teams in emerging countries with limited access to specialized tools.
- Combines software creation, commands, telemetry, sensor data, and AI-guided error explanations in one interface.
- Preparing deployment on affordable STM32 hardware widely used in Brazilian engineering and satellite projects.
- Paper accepted for presentation at AIAA SciTech 2027, an international aerospace conference.

Embedded Software for a CubeSat [Video] **Jan. 2024–May 2024**

- Built real-time software for a CubeSat prototype that autonomously oriented solar panels toward light.
- Integrated mission modes, attitude control, and power management on the physical prototype.
- Connected the satellite to a ground station and validated live command and telemetry exchange.

SKILLS

Programming: C, Python, Bash, MATLAB; familiar with C++, SystemC, VHDL, Verilog, and Assembly
Languages: Portuguese native, French C1, English B2, Spanish B1